UNIT-4. Design of Aircraft Structures:-

Design Criteria

Design Criteria in relation to aircraft design and in terms of mode of failure.

- Some of the important factors which are considered in modern a/c design are.
  - (a) long life
  - (b) fail safe
  - (c) fatigue
  - (d) Corrosion
  - (e) damage tolerant design.

NOTE: Although major focus in early development of aircrafts was stress analysis under the assumption that aircraft is rigid, modern a/c requirements are different.

The rigid a/c assumption is no longer valid because application of load causes certain distortions of airframe resulting in load distribution which are different from the initial load assumed. Hence derivation of equilibrium load under critical cases and consequent sizing of the pre-liminary structure are inter-related.

The whole issue of static and dynamic distortions is the effect of aero-elasticity.

Hence, the early design criteria is very arbitrary, where the aircraft was assumed to be rigid.

Given below is the list of design criterea specifying the associated mode of failure and the data to be considered.

Mode of failure.	Design Criterea	Allowable data.
1. Static Strength.	Structure must support Ultimate load without failure for 3 sec's.	static properties
2. Deformation.	Must be free from Aero-elastic instabilities.	Stiffness, mode shapes, Dynamic Stability.
3. Fatigue and Crack initiation	1. Fail safe structure for specified service life. 2. Safe life component must be crack free throughout service.	Fatigue properties
4. Residual static strength	1. Fail safe structure must support 80-100% limit load without catastropic failure. 2. Partial failure of mono-lithic structure.	static Properties Fracture toughness properties.
5. Crack growth of damaged structure.	1. For fail safe, inspection techniques and frequency of inspection to avoid catastrophic failure. 2. For safe life, inspection and frequency for replacement.	Crack growth properties and fracture toughness properties.

Design Criterea and Design life Criterea :-

Though the major focus of structural designers are Strength, there are other problems associated with design.

(a) Fail safety (b) Fatigue. (c) Corrosion (d) Aero-elasticity.
(e) maintainability (f) Reliability (g) Inspectability (h) Producability

Aircraft design Criterea can be approached in two ways.
(1) Design Criterea & (2) Design life Criterea.

Design Criterea :-

This is based on strength Criterea, based on maximum load which the aircraft is expected to meet atleast once in its life multiplied by 'FOS' to cater for in-accuracies in assumptions and Calculations. This often leads to arbitrary and conservative estimate of load & results in heavy structural members.

Design Life Criterea

High Computational Capabilities enables us now to take into account the actual environment to which the aircraft is exposed, its usage which may Consist of Cyclic loads, fluctuations in temperature and Corrosive environment.

All aircrafts of the Same class operating in different environments do not have the same life and hence, maintainance and repair requirements of every individual aircraft can be studied.

Analysis methods used to determine the life of the aircraft are:-

(1) Fatigue Analysis (2) Fracture Mechanic's.

Both methods have similarities in load input, but differs in assuming the flaw in structural member.

The initial Condition for fatigue is Unflawed where as in fracture mechanic's, it is a detectable Crack.

Fatigue is Complete when specimen fails and fracture toughness is no. of Cycles for the Crack to grow to a specified length.

<u>Wide spread fatigue damage</u>:-

Wide Spread fatigue damage (WSFD) is defined as the 'Simultaneous presence of Cracks at multiple structural Sites of Sufficient Size ~~and~~ because of which it Cannot meet damage tolerance requirements'.

Example: (1) Corrosion pitting in Skin, which leads to large number of micro-holes at close distances.

(2) Small Undetected multiple Cracks in several adjacent fasteners / rivet holes.

The multiple element damage requires Simultaneous development of fatigue Cracking in adjacent structural elements where as multiple site damage is development of simultaneous development of fatigue cracking in the same structural element so that Cracks may Coalesce from one large Crack and reach Critical length leading to Catastrophic failure or a very Small residual strength.

Some of the Susceptible structural components to wide spread fatigue damage / failure are:-

(a) longitudinal skin joints
(b) Circumferential skin joints
(c) Fuselage frames
(d) Area's in the proximity of pressure bulkheads
(e) Stringer to frame attachment.
(f) Surrounding Window structures.
(g) Chordwise splices in lifting surfaces
(h) Rib to skin attachment.

DESIGN PRINCIPLES:-

Due to Complexity of differing structural Synthesis of different elements, function, location and Consequence of failure of that particular structural element no single design principle is used. But the aerospace structural design principle can be broadly classified into these categories.

(1) Safe life
(2) Fail safe
(3) Damage tolerant design

Safe life :-

The Component must remain Crack free during its Service life. To ensure Safe life period of a structure, it must be designed to have a life much higher than the Safe life period.

Design life = Scatter factor × Safe life period.

The value of the scatter factor must obviously be higher than one, and its exact value depends on accuracy of available design information which are basically statistical.

Ex:- Most Combact aircraft are designed to have a safe life of a few thousand flying hours. But Critical Components are to be demonstrated to have a fatigue life 3.33 times the safe life of the aircraft.

A structure designed by the safe life Concept generally Contains a Single load path and inspectable Crack length is of the order of Critical Crack length. Consequent inspection intervals to monitor the Structure Cannot be defined.
A failure of any safe Component is assumed to lead to Complete failure of the structure. The structure is retired after safe life whether they are damaged or not.

A fatigue resistant design of safe life structure is either based on theory or experiment result. The obtained life is divided by Scatter factor to get the Safe life period
The safe life estimation Suffers from the following two aspects.

(1) Safety of life is not protected against manufacturing defect and maintanance induced defects

(2) Replacement after safe life period, irrespective of whether it is damaged or not.

Fail safe :-

The structure will support designated loads in spite of failure of any single member or partial damage to part of the structure. The fail safe approach is necessary as the failure of the structure may endanger the safety of the aircraft. Hence, fail safe Components are designed such that even if there is a failure, alternate path for flow of loads are provided. Ideally an aircraft structure designed according to fail safe principles can sustain damage and remain airworthy untill the damage is detected and repaired. Further the structure should be capable of taking additional loads if adjacent structures fail.

For this the structure is tested for 1.25 times the limit load factor under laboratory Condition. Since, fail safe design depends on detection of Crack, periodic inspection schedule must be worked out to determine period b/w inspections.

There are essentially three major drawbacks which should be addressed in fail safe approach.

(1) Life is determined under normal laboratory condition and hence do not account for variations in manufacturing process during actual production.

(2) Laboratory environment is not a representative of actual service environment variation

(3) Both safe life and fail safe do not maximize usage of structure as they have to be replaced whatever may be the residual strength left in the structure.

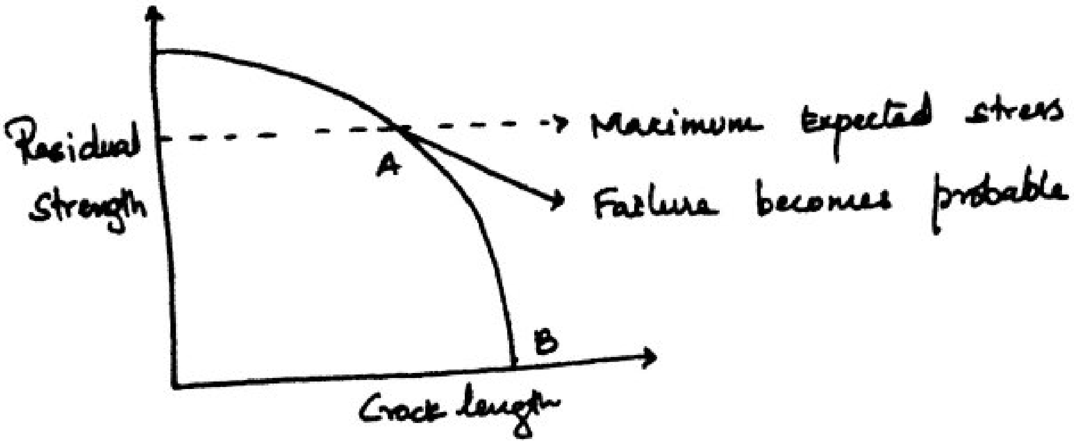
DAMAGE TOLERANT DESIGN :-

The damage tolerant design relies on inspection and repair to utilize the structure to the longest period of time. Damage tolerant design, generally is a very useful approach for designing a principal structure.

A principal structure is a structural member whose failure could result in the loss of aircraft.

By establishing inspection intervals for these elements based on the time it takes for a crack to grow from an initial detectable size to the critical length.

The crack growth behaviour can be precisely studied by following fracture mechanics and the residual strength left in the structure as the crack grows.



The residual strength of a Cracked Component decreases as the crack grows. The inspection schedule must be short enough to detect the crack before it reaches the point 'A'.

The loss of strength will be very high beyond AB.

The task of a designer is thus to design a structure to have a strength higher than A with a large crack length.

The damage tolerant design is a process which encompasses design, manufacture, use, inspection, and repair aspects together.

The fundamental basis for damage tolerant design are as follows.

(1). There is always some inherent flaw in the airframe, this flaw is Capable of growing even under loads lesser than yield strength. The Crack growth should be monitored such that it is detectable within inspection intervals. Critical length of a Crack is defined as, the length at which failure is sudden.

(2) There is always human negligence, during design Construction and operation of the Structure.

(3) Application of new materials / new manufacturing techniques are Capable of showing an Undesirable result at an unexpected time.

(4) Fundamental damage tolerant design depends on,

(1) Flaw and its growth into a Visible Crack & detection of it before it reaches Critical length.

(2) Incorporate design aspects to slow down Crack propagation. rate.

(3) Detection of growth within inspection interval under naked eye inspection.

(4) The Component should take the load with its residual strength until detection and repair.

(5) Damage tolerant design treats each and every individual aircraft as an individual, maintains its record from design to service, updates and predicts its life as Compared to fatigue Concept which tends to give life Common to all aircraft without Considering the manufacturing variations and damage each a/c has Suffered during its life time.

STRUCTURAL INTEGRITY

To ensure aircraft structure safety and long life, the aircraft material chosen should have the following desired characteristic's.

(1) Static strength efficiency.

(2) Fatigue strength

(3) Fracture toughness and crack growth rate.

(4) Availability and cost

(5) Fabrication characteristics

(6) Corrosion and embrittlement phenomenon.

(7) Compatibility with other materials

(8) Environmental stability.

The designer no longer chases a material solely on the basis of its mechanical strengths, but on proven ability to withstand minor damage and continue to serve without endangering safety of a/c before it is detected. This residual strength after damage is described as toughness is now supreme in choosing material for airframe.

The allowable value during design for all mechanical properties are further incorporated by assigning a factor of safety. The structural integrity is ensured by careful evaluation in all 70' links of design to manufacturing stage as described.

(1) Materials & Quality Control:-
Ductility, stress-strain, Homogenity, residual stress, heat treatment control, stress concentration, thermal stability, specification conformance, blue print conformance.

(2) Loads & Environment:-
Flight load, Ground load, flight dynamic load, launching, and landing dynamics, flight loads, interial loads, flexibility effect, repeated load spectrum.

(3) Allowables based on Importance of Component:-
On yielding, fracture, fatigue, wear, creep, deflections, stiffness and buckling.

(4) Stress Analysis:- Skin Panels, Beam Analysis, strain capability, stress concentration, Joint analysis, buckling analysis, fitting analysis, experimental stress analysis.

(5) Stiffness Criteria:- Flutter, Control system stability, Panel flutter, Vibrations, roll rate, divergence, dynamic responce to gust, gun firing, stores etc.

(6) Materials and Construction:- Fasteners, Welding, bonding, forging, Casting, extrusions, sheet metal, sandwich, bearings etc.

(7) Component Analysis:- Wing, tail, fuselage, thermal analysis, Deflection analysis, stiffness.

In order to ensure structural integrity, testing at component level and complete aircraft are undertaken to verify voracity of design.

Life Assessment procedure :-

Philosophy of fatigue life design :-

Fatigue life design

Fail Safe

(1) Structure has the Capability to contain fatigue and other damages.

(2) Achieved by :-

(a) Multiplicity of structural members
(b) Load transfer Capability b/w members
(c) Slow Crack propagation property
(d) Inspection Control.
(e) Fatigue problem reduced to maintaenance problem - Damage tolerant design.

Safe life.

(1) structure resist load within prescribed life.

(2) Achieved by.

(a) Knowledge of Varying loads and environmental effects
(b) Fatigue is treated as safety problem
(c) Fatigue life estimation by actual testing and applying Scatter factor.

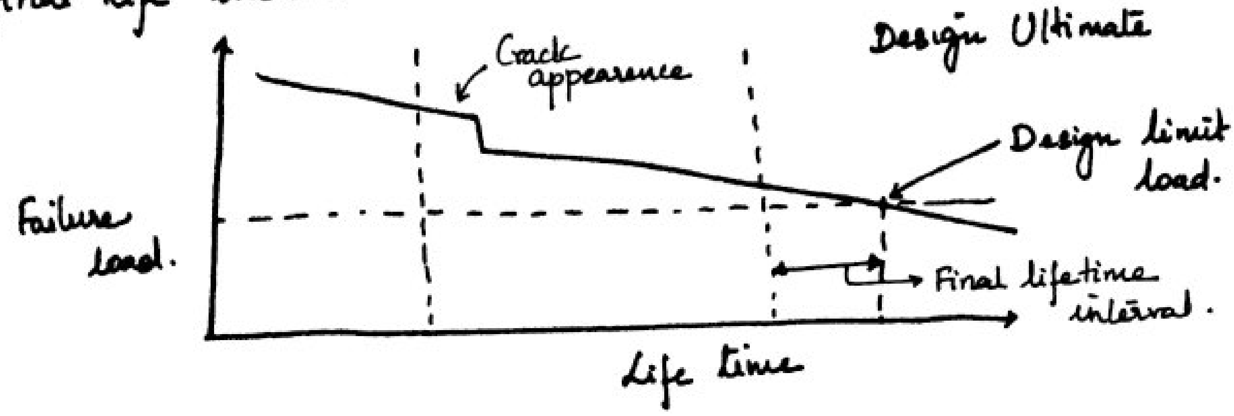
Fatigue is a progressive failure mechanism and the material degration initiates with (1) Load Cycle
(2) Environmental degradation.

The degradation accumulates and progresses and Culminates with Complete failure.

The life period from first cycle to complete failure can be divided into 3-stages.

(a) Initial life. (b) Post safe life period

(c) final life interval.



(a) Initial life Interval :-

Initial life interval during which a complete failure can occur only when the applied load exceeds the design ultimate strength. This time interval is generally refered to as fatigue life or safe life interval.

(b) Post safe life interval :-

Life interval after safe life interval during which a load equal or less than ultimate load can initiate a crack. The crack growth is a function of material fracture toughness properties.

(c) Final life interval :

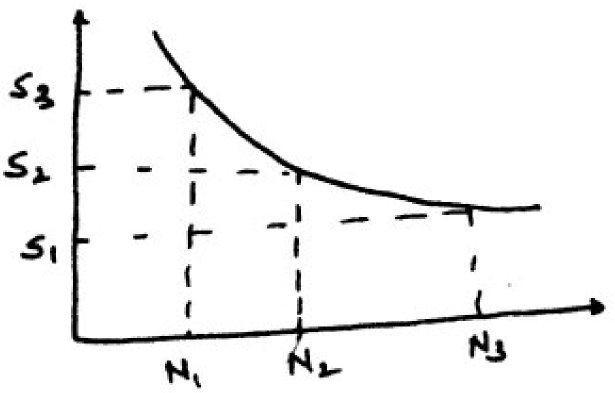
During which a complete failure will occur even when the applied load is below ultimate load.

Structural Life Estimation - Palmgren-Miner Method

The palm-gran minor method proposes that a fraction of fatigue life is used up in service when a load is applied. This fraction is defined as $\frac{n_i}{N_i}$ and remaining life is $1 - \frac{n_i}{N_i}$

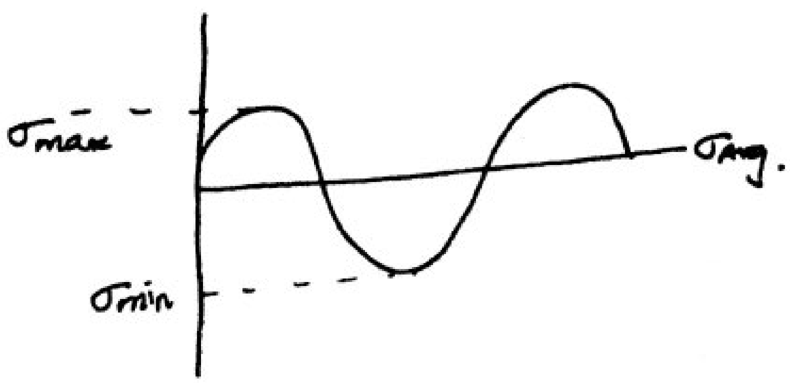
→ The fatigue damage incurred at a given stress level is proportional to the number of cycles applied at that stress level n_i divided by the total number of cycles required to cause failure at the same level.

$$\sum_{i=1}^{k} \frac{n_i}{N_i} = 1.0 \text{ at failure}$$, where k is the no. of stress levels considered.



Life Assessment procedures - Analysis Method.

Consider a body subjected to an oscillatory stress between σ_{max} and σ_{min} as shown.



The stress intensity factor K is a function of Stress. Hence, when a Cracked body is subjected to an oscillatory load, the stress intensity factor also oscillates b/w K_{max} and K_{min} with an average value.

$$K_{mean} = \frac{K_{max} + K_{min}}{2}$$

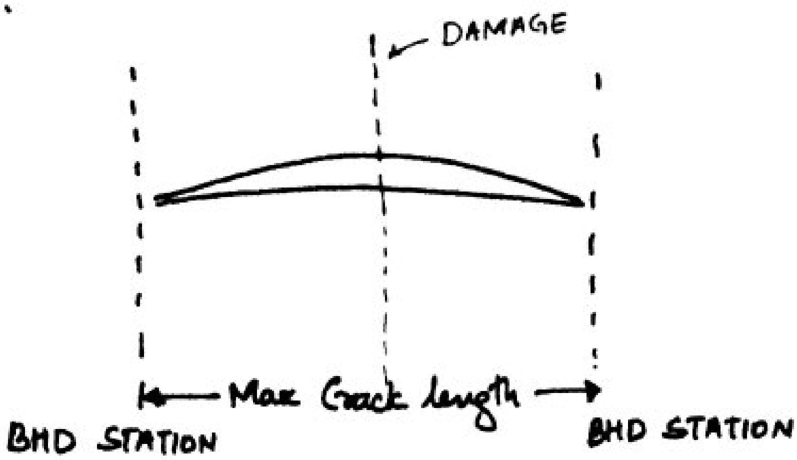
In order to maintain the Structural integrity throughout the economic Service life of the aircraft it should retain Sufficient residual strength in the presence of fatigue Cracks

Most Critical of the failure are at the site of the bulkhead This can be caused by two reasons.

1. Longitudinal and Circumferential lap or butt splices with rivets under pressure loading forming a Single dominant Crack at the bulk head station.
2. Damage Caused by external Source at a bulkhead station.

The Structural integrity requirement Under Such failure is that fuselage structure must be Capable of arresting a 2-bay Crack. When damage occurs at a bulkhead the Crack Can grow but should stop at the bulkheads adjacent to it. Hence, the maximum Crack length is equal to twice the bulkhead spacing and no more Under Such Cracks the fuselage should retain Sufficient residual strength to complete the mission and return safely.

This 2-bay Crack is now an aircraft industry standard for development of wide body passenger aircrafts.



Stress intensity factor K determines the stress field around the crack tip in a linear elastic material for a remotely applied normal stress.

$K = \alpha \sigma \sqrt{\pi a}$ where α - functionality Constant

σ - applied stress

or $K_{Ic} = Y \sigma \sqrt{\pi \cdot a}$ $2a$ - Crack length.

The rate of Crack growth da/dn is a function of mean stress intensity factor K_m and $\Delta K = K_{max} - K_{min}$

$$\frac{da}{dn} = f(\Delta K, K_m)$$

The functional relation b/w them are found to be

$$\log \frac{da}{dn} = \log c + n \log \Delta K$$ for given stress

$$\frac{da}{dn} = C \Delta K^n$$ where C and n are material dependent.

Design life Criteria:

When stress level oscillates b/w σ_{max} and σ_{min} the rate of Crack da/dn is a function of $\Delta k = k_{max} - k_{min}$

Given $k = \frac{2}{\pi}\sigma\sqrt{\pi a}$, $\Delta k = \frac{2}{\pi}\Delta\sigma\sqrt{\pi a}$

The relation b/w Crack growth rate and Δk is given by

$$da/dn = C\Delta k^n.$$

$$\frac{da}{dn} = C\left[\frac{2}{\pi}\Delta\sigma\sqrt{\pi a}\right]^n = C\left[\frac{2}{\sqrt{\pi}}\Delta\sigma\right]^n a^{n/2}$$

Integrating for Crack from a_0 to a

$$\int_{a_0}^{a}\frac{da}{a^{n/2}} = \int_0^N C\left[\frac{2}{\sqrt{\pi}}\cdot\Delta\sigma\right]^n\cdot dn$$

$$\left[\frac{a^{1-n/2}}{1-n/2}\right]_{a_0}^{a} = C\left[\frac{2}{\sqrt{\pi}}\cdot\Delta\sigma\right]^n\cdot N.$$

Hence, $$N = \frac{a^{1-n/2} - a_0^{1-n/2}}{C\left[\frac{2}{\sqrt{\pi}}\Delta\sigma\right]^n\left[1-n/2\right]}.$$

At Critical radius a_c, the Crack propogation to Catastrophic failure and Corresponding cycles N_f is the number of Cycles to failure.

$$N_f = \frac{a_c^{1-n/2} - a_0^{1-n/2}}{C\left[\frac{2}{\sqrt{\pi}}\Delta\sigma\right]^n\left[1-n/2\right]}.$$

2-Bay Crack Criteria :-

Transport aircrafts Cruising at high altitudes require pressurization of the Cabin.
The fuselage structure is then subjected to one pressurization Cycle per flight. For fuselages this pressure cycling is the most critical loads that create fuselage Crack. Furthermore this fuselage may be damaged due to bird hit or a loose Component hitting the fuselage.
Ex:- A broken propellar hitting the side of the fuselage.

In Order for the Crack not to propogate beyond twice the bulkhead station the following Conditions need to be established.

(1) $K < K_c$ stress intensity factor K should be less than K_c, a Critical Value of K beyond which the growth of Crack will Continue.

(2) The rivets adjacent to damage site should be intact.
(3) The bulk-heads are intact.

Weight- prediction Methods :-

→ Importance of Weight prediction :-
Most performance characteristics are adversely affected by increasing weight and improved by decreasing weight. Weight data is a tool used by engineering management as well as potential customers in arriving at good design decisions.

Weight Control can be described as the process by which lightest possible airplanes is derived within the constraints of governing design Criteria.

Weight prediction Methods :-

Preliminary estimation of weight is one of the first steps in the evolution of a new airplane design. The preliminary weight available power and required power and performance are basic parameters which determine the size and general configuration of the airplane.

(a) Wing Weight Estimation :-

Usually bending moment in flight is assumed to be decisive for most of the primary structure. Torsional stiffness & safe guard against flutter may amount to 20% of wing weight.

(b) Tail Weight Estimation :-

The tail weight is only a small part of the (MTOW) Maximum take-off weight, which is about 2%. If the taiplane area is not yet known, then the total tail plane weight may be assumed about 4% of the (OEW) Operating Empty Weight.

(c) Fuselage Weight Estimation :-

The fuselage includes the complete shell, excluding the nose radome but including wind shield, windows, doors, flooring, supports, & Cargo Compartments.

(d) Landing Gear Estimation:-

The weight of the Conventional landing gears may be found by Summation of the main gear and nose gear each predicted Separately. In many transport aircrafts, the critical load is formed by the landing impact load and the maximum landing weight. (MLW)
A reasonable approximation of the weight of retractable landing gears is 5% of the maximum landing weight.

(e) Engine Nacelle and Mounting Weight Estimation:-

(1) For aircrafts with turbo-prop engines.

$W_n = 0.14$ lb per take off of Equivalent Shaft horse-power.

(2) For aircrafts with pod-mounted turbofan or turbo-jet:-

$$W_n = 0.055\, T_{TO}$$

T_{TO} = Maximum thrust at airplane take off.

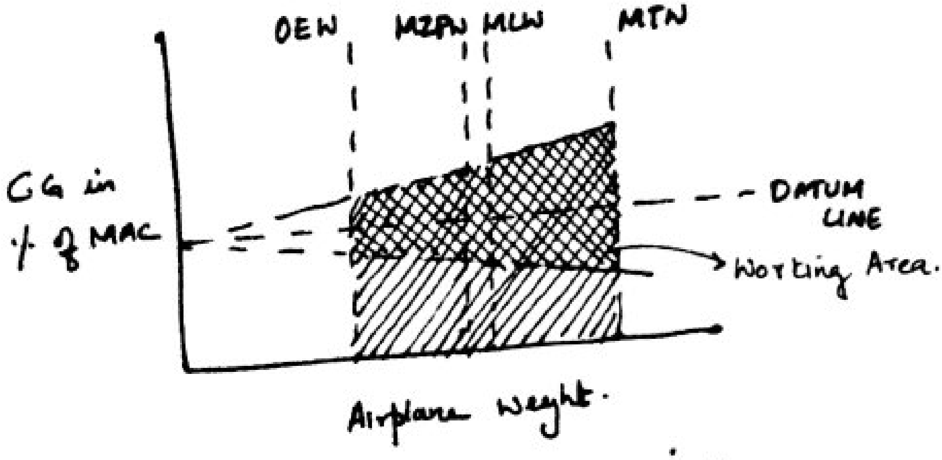
(3). For aircrafts with high by-pass turbo-fans with short-fan duct:-

$$W_n = 0.065\, T_{TO}.$$

BALANCE & LOADABILITY

Airplane loading flexibility is the degree of freedom afforded by the airline with useful load combinations that they actually encounter during daily operations.

- The balance grid depicted is used during the whole life span of an airplane from Concept to retirement.



where OEW – Operating Empty weight
MZFW - Maximum Zero fuel weight
MLW - Maximum Landing weight
MTW - Maximum ~~[illegible]~~ Taxi weight.

If the weight and moment coordinates of a balance grid start at zero then the working area of the grid becomes far to small for practical use.

<u>Numericals</u>:-

Suppose a wing component is made from a material with strain fracture toughness of 40 Mpa$\sqrt{m}$. and maximum internal crack length of 4.0mm. Given that fracture results at a stress of 365 Mpa. when max internal crack is 2.5mm. Compute stress level at which fracture will occur for a critical internal crack length of 4.0mm.

<u>Sol</u>:: First solve for Y from. $Y = \frac{K_{IC}}{\sigma\sqrt{\pi . a}}$ 2a = Critical crack length

$$= \frac{40 \text{ Mpa}\sqrt{m}}{365\sqrt{\pi\left(\frac{2.5\times10^{-3}}{2}\right)}} = 1.76$$

Now, we solve for σ_c, rearranging the equation:-

$$\sigma_c = \frac{K_{Ic}}{Y\sqrt{\pi a}} = \frac{40\ \text{Mpa}\sqrt{m}}{1.76\sqrt{\pi\left(\frac{4\times10^{-3}}{2}\right)}} = 288\ \text{Mpa}.$$

Problem 2:- An Aluminium alloy has the following fatigue characteristics.

Stress 1, $N_1 = 45$ cycles. $n_1 = 5$ cycles
Stress 2, $N_2 = 310$ cycles $n_2 = 60$ cycles
Stress 3, $N_3 =$ ~~4995~~ 12,400 cycles $n_3 = 495$ cycles.

How many times can the following sequence be repeated?

Sol. From Miner's Rule, we have.

$$\text{Damage} = \sum_i \frac{n_i}{N_i} = \frac{n_1}{N_1} + \frac{n_2}{N_2} + \frac{n_3}{N_3}$$

$$\text{Damage} = \frac{5}{45} + \frac{60}{310} + \frac{495}{12400} = 0.345.$$

The life of the component is entirely exhausted when damage is 1. At this point the sequence is repeated X times to make the damage 1.

$$X(0.345) = 1, \text{ hence } X = 2.9 \text{ times}.$$

Once 2.9 times is complete, the damage becomes 1.

Problem 3

A structure is subjected to a pulsating loading, $\sigma_{min} = 0$, Embedded Circular Cracks are discovered in the structure. The Crack diameters are $2a_0$, which is much smaller than the thickness of the material. It follows paris law.

(a) Determine largest stress range $\Delta\sigma_0$, the structure may be subjected to, if crack propagation is not allowed. $\alpha = 0.637$.

(b) Determine the maximum stress range that may be allowed if the structure is loaded by $N = 10{,}000$ cycles. During the loading the crack is not allowed to grow so large that it becomes unstable.
FOS = 1.5

(B) Given Data:- $a_0 = 0.0005m$ $\sigma_y = 620$ Mpa $\Delta K_{TH} = 2MN/m^{3/2}$
$K_{IC} = 36\ MN/m^{3/2}$ $n = 4.0$ and $C = 1.12\times10^{-11} m^7/mN^4$.

Sol:- (a) No crack propagation is allowed, hence, ΔK_I should be less than ΔK_{TH}. Hence,

$$K_I = \sigma_0 \sqrt{\pi a}\,(0.637)$$

or $$\Delta K_I = \Delta\sigma_0 \sqrt{\pi . a}\,(0.637)$$

when no crack propagation, then, $\Delta K_I \leq \Delta K_{th}$

$$\Delta\sigma_0 \sqrt{\pi a}\,(0.637) \leq \Delta K_{TH}.$$

Substituting value of ΔK_{TH} and solving for $\Delta\sigma_0$,

$$\Delta\sigma_0 = 79 \text{ Mpa}.$$

Hence, we can say that, if $\Delta\sigma_0 \leq 79$ mpa no crack growth is expected.

(b) Crack propagation :-

$N = 10^4$ cycles.

Similiarly $K_I = 0.637\,\sigma_0 \sqrt{\pi a}$.

When LEFM is valid then $K_{I\,max} = \frac{K_{IC}}{S}$

hence = $0.637\sigma_0 \sqrt{\pi a_c} = \frac{K_{IC}}{S}$, where S is 'fos'

rearranging and solving, for a_c

$$a_c = \frac{1}{\pi}\left[\frac{K_{IC}}{0.637\,\sigma_0\,S}\right]^2 = \frac{452}{\sigma_0^2}$$

here σ_0 is equal to stress range because $\sigma_{min} = 0$.

Now, we know a_0 and a_c. During N cycles the crack will grow from a_0 to a_c. hence by paris law,

$$\frac{da}{dN} = C\,(\Delta K_I)^n = C\,(0.637\,\Delta\sigma_0 \sqrt{\pi a})^n$$

Integrating.
$$\int_{a_0}^{a_c} \frac{da}{(\sqrt{a})^n} = \int_0^N C\,(0.637\,\Delta\sigma_0 \sqrt{\pi})^n\, dN.$$

after evaluating integrals.
$$\frac{a_c^{1-n/2} - a_0^{1-n/2}}{1-n/2} = C\,(0.637\,\Delta\sigma_0\sqrt{\pi})^n \cdot N.$$

Substituting, we get
$$\frac{\left(452/(\Delta\sigma_0)^2\right)^{-1} - (0.5\times10^{-3})^{-1}}{-1} = 1.2\times10^{-11}\,(0.637\,\Delta\sigma_0\sqrt{\pi})^4 \cdot \times10^4$$

Solving for $\Delta\sigma_0$, we get $\Delta\sigma_0 = 324\text{ Mpa} = \sigma_0$.

hence $a_c = \frac{452}{324^2} = 0.0043\,m$. Hence stress range should be b/w 79 mpa & 324 Mpa.